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Seasoning Model Soups for Robustness to Adversarial and Natural Distribution Shifts

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Abstract

Adversarial training is widely used to make classifiers robust to a specific threat or adversary, such as ℓ_p-norm bounded perturbations of a given p-norm. However, existing methods for training classifiers robust to multiple threats require knowledge of all attacks during training and remain vulnerable to unseen distribution shifts. In this work, we describe how to obtain adversarially-robust model soups *(i.e., linear combinations of parameters) that smoothly trade-off robustness to different ℓ_p-norm bounded adversaries. We demonstrate that such soups allow us to control the type and level of robustness, and can achieve robustness to all threats without jointly training on all of them. In some cases, the resulting model soups are more robust to a given ℓ_p-norm adversary than the constituent model specialized against that same adversary. Finally, we show that adversarially-robust model soups can be a viable tool to adapt to distribution shifts from a few examples.*

1. Introduction

Deep networks have achieved great success on several computer vision tasks and have even reached super-human accuracy [16, 28]. However, the outputs of such models are often brittle, and tend to perform poorly on inputs that differ from the distribution of inputs at training time, in a condition known as *distribution shift* [34]. Adversarial perturbations are a prominent example of this condition: small, even imperceptible, changes to images can alter predictions to cause errors [2, 41]. In addition to adversarial inputs, it has been noted that even natural shifts, e.g. different weather conditions, can significantly reduce the accuracy of even the best vision models [11, 18, 36]. Such drops in accuracy are undesirable for robust deployment, and so a lot of effort has been invested in correcting them. Adversarial training [31] and its extensions [13, 35, 51] are currently the most effective methods to improve empirical robustness to adversarial attacks. Similarly, data augmentation is the basis of several techniques that improve robustness to non-adversarial/natural shifts [3, 9, 19]. While significant progress has been made on defending against a specific, selected type of perturbations (whether adversarial or natural), it is still challenging to make a single model robust to a broad set of threats and shifts. For example, a classifier adversarially-trained for robustness to ℓ_p-norm bounded attacks is still vulnerable to attacks in other ℓ_q-threat models [26, 42]. Moreover, methods for simultaneous robustness to multiple attacks require jointly training on all [30, 32] or a subset of them [6]. Most importantly, controlling the trade-off between different types of robustness (and nominal performance) remains difficult and requires training several classifiers.

Inspired by *model soups* [46], which interpolate the parameters of a set of vision models to achieve state-of-the-art accuracy on IMAGENET, we investigate the effects of interpolating robust image classifiers. We complement their original recipe for soups by own study of how to pre-train, fine-tune, and combine the parameters of models adversarially-trained against ℓ_p-norm bounded attacks for different p-norms. To create models for soups, we pre-train a single robust model and fine-tune it to the target threat models (using the efficient technique of [6]). We then establish that it is possible to smoothly trade-off robustness to different threat models by moving in the convex hull of the parameters of each robust classifier, while achieving competitive performance with methods that train on multiple p-norm adversaries simultaneously. Unlike alternatives, our soups can uniquely *(1)* choose the level of robustness to each threat model without any further training and *(2)* quickly adapt to new unseen attacks or shifts by simply tuning the weighting of the soup.

Previous works [21, 27, 47] have shown that adversarial training with ℓ_p-norm bounded attacks can help to improve performance on natural shifts if carefully tuned. We show that model soups of diverse classifiers, with different types of robustness, offer greater flexibility for finding models that perform well across various shifts, such as IMAGENET variants. Furthermore, we show that a limited number of images of the new distribution are sufficient to select the

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weights of such a model soup. Examining the composition of the best soups brings insights about which features are important for each dataset and shift. Finally, while the capability of selecting a model specific to each image distribution is a main point of our model soups, we also show that it is possible to jointly select a soup for average performance across several IMAGENET variants to achieve better accuracy than adversarial and self-supervised baselines [15, 21].

Contributions. In summary, we show that soups

- can merge nominal and ℓ_p-robust models (for various p): efficient fine-tuning from one robust model obtains a set of models with diverse robustness [6] and compatible parameters for creating model soups [46] (Sec. 3),
- can control the level of robustness to each threat model and achieve, without more training, competitive performance against multi-norm robustness training (Sec. 4),
- are not limited to interpolation, but can find more effective classifiers by extrapolation (Sec. 4.3),
- enable adaptation to unseen distribution shifts on only a few examples (Sec. 5).

2. Related Work

Adversarial robustness to multiple threat models. Most methods focus on achieving robustness in a single threat model, i.e. to a specific type of attacks used during training. However, this is not sufficient to obtain robustness to unseen attacks. As a result, further work aims to train classifiers for simultaneous robustness to multiple attacks, and the most popular scenario considers a set of ℓ_p-norm bounded perturbations. The most successful methods [30, 32, 42] are based on adversarial training and differ in how the multiple threats are combined. Notably, all need to use every attack at training time. To reduce the computational cost of obtaining multiply-robust models, one can fine-tune a singly-robust model by one of the above mentioned methods [6], even for only a small number of epochs. Our soups are more flexible by skipping simultaneous adversarial training across multiple attacks.

Adversarial training for distribution shifts. While robustness to adversarial attacks and natural shifts are not the same goal, previous works nevertheless show that it is possible to leverage adversarial training with ℓ_p-norm bounded attacks to improve performance on the common corruptions of [18]. First, AdvProp [47] co-trains models on clean and adversarial images (in the ℓ_∞-threat model) with dual normalization layers that specialize to each type of input. The clean branch of the dual model achieves higher accuracy than nominal training on IMAGENET and its variants. Similar results are obtained by Pyramid-AT [21] by its design of a specific attack to adversarially train vision transformers. Finally, [27] carefully selecting the size of the adversarial perturbations, i.e. their ℓ_∞- or ℓ_2-norm, for standard adversarial training [31] achieves competitive performance on common corruptions on CIFAR-10 and IMAGENET-100.

Model soups. Ensembling or averaging the parameters of intermediate models found during training is an effective technique to improve both clean accuracy [22, 25] and robustness [12, 35]. Recently, [46] propose *model soups* which interpolate the parameters of networks fine-tuned with different hyperparameters configurations from the same pre-trained model. This yields improved classification accuracy on IMAGENET. Along the same line, [24] fine-tune a model trained on IMAGENET on several new image classification datasets, and show that interpolating the original and fine-tuned parameters yields classifiers that perform well on all tasks.

3. Model Interpolation across Different Tasks

In the following, we formally introduce the two main components of our procedure to merge adversarially robust models: *(1)* obtaining models which can interpolated by fine-tuning a single ℓ_p-robust classifiers, and *(2)* interpolation of their weights to balance their different types of robustness. We highlight that our setup diverges from that of prior works about parameters averaging: in fact, both [24, 46] combine models fine-tuned on the same task, i.e. achieving high classification accuracy of unperturbed images, either on a fixed dataset and different hyperparameter configurations [46], or varying datasets [24]. In our case, the individual models are trained for robustness to different types of attacks, i.e. with distinct loss functions.

3.1. Adversarial training and fine-tuning

Let us denote $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_i$ the training set, with $\boldsymbol{x}_i \in \mathbb{R}^d$ indicating an image and $y_i \in \{1, \dots, K\}$ the corresponding label, and $\Delta : \mathbb{R}^d \to \mathbb{R}^d$ the function which characterizes a threat model, that maps an input $\boldsymbol{x}$ to a set $\Delta(\boldsymbol{x}) \subset \mathbb{R}^d$ of possible perturbed versions of the original image. For example, ℓ_p-norm bounded adversarial attacks with budget $\epsilon > 0$ in the image space can be described by

$$\Delta(\boldsymbol{x}) = \{\boldsymbol{\delta} \in \mathbb{R}^d : \|\boldsymbol{\delta}\|_p \leq \epsilon, \boldsymbol{x} + \boldsymbol{\delta} \in [0,1]^d\}. \quad (1)$$

Then, one can train a classifier $f : \boldsymbol{\theta} \times \mathbb{R}^d \to \mathbb{R}^K$ parameterized by $\boldsymbol{\theta} \in \Theta$ with adversarial training [31] by solving

$$\min_{\boldsymbol{\theta} \in \Theta} \sum_{(\boldsymbol{x}, y) \in \mathcal{D}} \max_{\boldsymbol{\delta} \in \Delta(\boldsymbol{x})} \mathcal{L}(f(\boldsymbol{\theta}, x + \boldsymbol{\delta}), y), \quad (2)$$

for a given loss function $\mathcal{L} : \mathbb{R}^K \times \mathbb{R}^K \to \mathbb{R}$ (e.g., cross-entropy), with the goal of obtaining a model robust to the perturbations described by Δ. Note that this boils down to nominal training when $\Delta(\cdot) = \{\mathbf{0}\}$ and no perturbation is applied on the training images. We are interested in the case where multiple threat models are available, and indicate with Δ_{nominal} nominal training, and Δ_p for $p \in \{\infty, 2, 1\}$

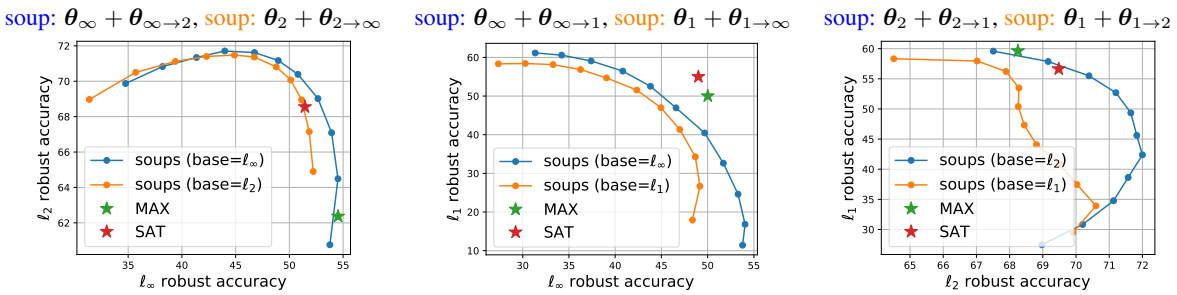

Figure 1. **Soups of two models on CIFAR-10:** for all pairs (p, q) we show the ℓ_p- vs ℓ_q-robust accuracy of the soups $w\cdot\boldsymbol{\theta}_p+(1-w)\cdot\boldsymbol{\theta}_{p\to q}$ and $w\cdot\boldsymbol{\theta}_q+(1-w)\cdot\boldsymbol{\theta}_{q\to p}$ varying $w\in[0,1]$. We also show results for MAX and SAT with simultaneous use of the two threat models.

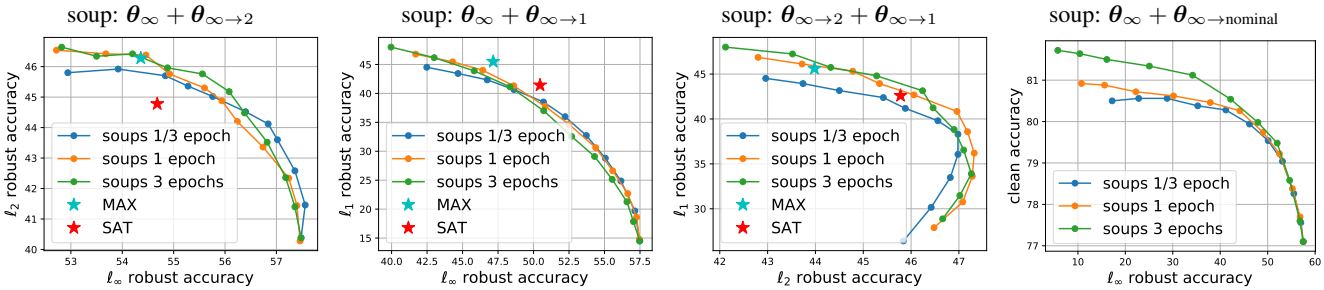

Figure 2. **Soups of two models on IMAGENET:** for $p\in\{2,1\}$ we show the ℓ_p- vs ℓ_∞-robust accuracy of the soups $w\cdot\boldsymbol{\theta}_\infty+(1-w)\cdot\boldsymbol{\theta}_{\infty\to p}$ varying $w\in[0,1]$ (first and second columns). Moreover, we show the soups obtained combining $\boldsymbol{\theta}_{\infty\to 2}$ and $\boldsymbol{\theta}_{\infty\to 1}$ (third), or $\boldsymbol{\theta}_\infty$ and $\boldsymbol{\theta}_{\infty\to\text{nominal}}$ (fourth). For the case of two ℓ_p-threat models, we also show the results of fine-tuning models with MAX and SAT.

the perturbations with bounded ℓ_p-norm as described in Eq. 1. We focus on such tasks since they are the most common choices for adversarial defenses, in particular by methods focusing on multiple norm robustness [32, 42].

Notably, it is possible to efficiently obtain a model robust to Δ_q by fine-tuning for a single (or few) epoch with adversarial training w.r.t. Δ_q a classifier pre-trained to be robust in Δ_p, with $p\neq q$ and $p,q\in\{\infty,2,1\}$ [6]. For example, in this way one can efficiently derive specialized classifiers $\Delta_2,\Delta_1,\Delta_{\text{nominal}}$ for each task from a single model f robust w.r.t. ℓ_∞. However, this does not work well when a nominal classifier is used as starting point for the short fine-tuning. In the following, we denote with $\boldsymbol{\theta}_{p\to q}$ the parameters resulting from fine-tuning to Δ_q a base model $\boldsymbol{\theta}_p$.

3.2. Merging different types of robustness via linear combinations in parameter space

We want to explore the properties of the models obtained by taking linear combinations of the parameters of classifiers with different types of robustness. To do so, there needs to be a correspondence among the parameters of different individual networks: [46] achieve this by merging differently fine-tuned versions of the same pre-trained model. As mentioned above, fine-tuning an ℓ_p-robust classifier allows to change it to achieve robustness in a new threat model [6]: we exploit such property to create *model soups*, as named by [46]. Formally, we create a model soup from n individual networks with parameters $\boldsymbol{\theta}^1,\dots,\boldsymbol{\theta}^n$ with weights $\boldsymbol{w}=(w_1,\dots,w_n)\in\mathbb{R}^n$ as

$$\boldsymbol{\theta}^{\boldsymbol{w}}=\sum_{i=1}^{n} w_i\cdot\boldsymbol{\theta}^i, \tag{3}$$

and the corresponding classifier is given by $f(\boldsymbol{\theta}^{\boldsymbol{w}},\cdot):\mathbb{R}^d\to\mathbb{R}^K$. While any choice of $\boldsymbol{w}$ is possible, we focus on the case of affine combinations, i.e. $\sum_i w_i=1$. Moreover, we consider soups which are either convex combinations of the individual models, with $w_1,\dots,w_n\geq 0$, or obtained by extrapolations, i.e. with negative elements in $\boldsymbol{w}$.

4. Soups for ℓ_p-robustness

We measure adversarial robustness in the ℓ_p-threat model with bounds ϵ_p: on CIFAR-10 we use $\epsilon_\infty=8/255$, $\epsilon_2=128/255$, $\epsilon_1=12$, on IMAGENET $\epsilon_\infty=4/255$, $\epsilon_2=4$, $\epsilon_1=255$. If not specified otherwise, we use the full test set for CIFAR-10 and 5000 images from the IMAGENET validation set, and attack by AUTOPGD [4, 5] with 40 steps. More details are provided in App. A.